

as if each pair constituted a single larger stratum. This *collapsed strata* technique leads to some overestimation of sampling errors, but providing the paired strata are closely similar, the overestimation should not be serious. The collapsing of pairs of strata, each with a single primary selection, sets up a design with two sampled PSUs in each stratum, a design commonly known as a *paired selection design*. The PSUs in the collapsed strata are usually treated as having been sampled with replacement so that the simple with-replacement variance estimators described earlier can be used.

6. Probability Proportional to Size Sampling

The last section assumed that the clusters were of equal size, but this is rarely so in practice. The natural groupings that the sampler takes advantage of to serve as clusters almost always vary in size, often in a major way. The classes in a high school will not all contain 24 students, but may perhaps vary between 20 and 30; city blocks vary much more in the number of households they contain, as also do counties (which are frequently used as PSUs for national samples in the United States). The difficulties this variation in size creates will now be explained and methods will be described by which these difficulties can be overcome.

For ease of exposition we will use an artificially small example involving the selection of an epsem sample of dwellings (the elements) from a set of 9 city blocks (the PSUs); the example might be thought of as one stratum in a much larger design. The 9 blocks contain 315 dwellings in total, and a sample of 21 is desired, implying an overall sampling fraction of 1 in 15. At the first stage of sampling, 3 blocks are to be drawn, and then dwellings are sampled within the selected blocks. Initially we assume that the sizes of the blocks—i.e., the numbers of dwellings they contain—are known without error. These sizes, with B_{α} denoting the number of dwellings in block α , are as follows:

Block:	1	2	3	4	5	6	7	8	9	Total
B_{α}	20	100	50	15	18	43	20	36	13	315

The first design to be considered takes an SRS of three blocks at the first stage. Each block thus has a probability of $3/9 = 1/3$ of being selected. The selection probability for a dwelling can be obtained using

the multiplication rule of probabilities. In general, with a two-stage design the probability of element i in cluster j appearing in the sample is

$$P(\alpha\beta) = P(\alpha)P(\beta|\alpha) \quad [19]$$

where $P(\alpha)$ is the probability of cluster j being selected, and $P(\beta|\alpha)$ is the probability of element i in cluster j being chosen at the second stage, given that cluster j was selected at the first stage. This equation, which can be extended to cover more sampling stages when necessary, is sometimes known as the *selection equation* in the survey sampling literature.

The current example requires an overall epsem design with $f = P(\alpha) = 1/15$. With the clusters being selected with equal probabilities, $P(\beta|\alpha) = 1/3$, it then follows from the selection equation that $P(\alpha\beta) = 1/5$. In other words, the sampling rate at the second stage within each selected block is $1/5$. Consider now some possible samples that could arise from this sample design. At one extreme the sampled blocks could comprise the three smallest blocks—4, 5, and 9—while at the other extreme they could comprise the three largest blocks—2, 3, and 6. In the former case, the application of the 1 in 5 sampling fraction within the sampled blocks would yield a total sample of about 9 dwellings, and in the latter case it would yield a total sample of about 39. On average, over all possible simple random samples of three blocks that could be selected, the sample size is 21, but the actual sample size achieved in a single application may deviate substantially from this figure.

The substantial variability in possible sample size in this example is partially accounted for by the unrealistically small sample of clusters being taken. On the other hand, in other sampling problems the variability in cluster sizes may be much greater than in the blocks considered here. It is clear that some control on the potential variability in sample size is needed. While the sample size rarely needs to be specified exactly, it does need to be kept within reasonable bounds.

One way to reduce the potential variability in sample size is to stratify the clusters by size. In this example the blocks could be divided into three strata according to size: One stratum would contain blocks 2, 3, and 6, the second would contain blocks 1, 7, and 8, and the third blocks 4, 5, and 9. Selecting one block from each stratum would reduce the variability in sample size to extremes from about 15 (blocks 1 or 7, 6, and 9) to 31 (blocks 2, 5, and 8). With a larger number of selections,

stratification by size usually achieves adequate control on sample size. However, the use of size stratification reduces the amount of use that can be made of other stratification factors. For this reason, an alternative procedure for controlling sample size, now to be described, is generally preferred.

Let us specify the conditions that we would like the sample to satisfy in the example: (1) It should be an epsem sample, (2) it should be restricted to three blocks, and (3) the sample size should ideally be fixed at $n = 21$, no matter which blocks are selected. The first and third conditions imply that $P(\alpha) = 1/15$ in the selection equation. The second and third conditions are met if a sample of 7 dwellings is selected from each of three selected blocks, irrespective of the blocks' sizes; if this is done, the selection probability at the second stage in sampled block α is $P(\beta|\alpha) = 7/B_\alpha$. Substituting $P(\alpha) = 1/15$ and $P(\beta|\alpha) = 7/B_\alpha$ in the selection equation gives

$$\frac{1}{15} = P(\alpha) \cdot \frac{7}{B_\alpha}$$

which may be solved to give the probability of selection of blocks α as $P(\alpha) = B_\alpha/105$. Thus, all three conditions are met if the blocks are sampled with probability proportional to their B_α 's, i.e., with *probability proportional to size* (PPS).

In general, the selection equation for an epsem two-stage sample with PPS selection is given by

$$P(\alpha\beta) = f = \frac{n}{N} = \left(\frac{aB_\alpha}{\sum B_\alpha} \right) \left(\frac{b}{B_\alpha} \right) \quad [20]$$

where a PSUs are selected by PPS, b elements are sampled from each selected PSU, $n = ab$, and $N = \sum B_\alpha$. The equation can be extended to a three-stage design, with a PSUs selected, b second stage units (SSUs) sampled within each selected PSU, and c elements sampled within each SSU as

$$P(\alpha\beta\gamma) = f = \frac{n}{N} = \left(\frac{aB_\alpha}{\sum B_\alpha} \right) \left(\frac{bB_{\alpha\beta}}{B_\alpha} \right) \left(\frac{c}{B_{\alpha\beta}} \right)$$

where now $n = abc$, $B_{\alpha\beta}$ is the size of SSU β in PSU α , and $\sum_\beta B_{\alpha\beta} = B_\alpha$.

The selection of the blocks with PPS can be achieved by cumulating their sizes as follows:

Block:	1	2	3	4	5	6	7	8	9
B_{α} :	20	100	50	15	18	43	20	36	13
Cumulative B_{α} :	20	120	170	185	203	246	266	302	315

Using the cumulative totals, numbers are associated with each block: Block 1 is allocated the 20 numbers from 001 to 020; block 2 the 100 numbers from 021 to 120; block 3 the 50 numbers from 121 to 170; and so on. In this way each block is assigned as many numbers as its size B_{α} . A random number chosen from 001 to 315 then selects a block with PPS. If the random number were, say 197, block 5 would be selected.

Three random numbers could be drawn in the manner outlined above to give the required 3 sampled blocks, but this with-replacement sampling scheme gives a block a chance of being selected more than once. Systematic sampling can be used to give a simple without-replacement PPS sampling method. The total size, 315, is divided by the number of selections to be made—3—to give the sampling interval of 105. A random number up to 105 is chosen, say 047, to determine the first selection, block 2. Then 105 is added to the random number to give 152, making block 3 the second selection; adding 105 again gives 257, making block 7 the third selection.

The ultimate cluster sampling approximation described in the last section can also be used with a PPS design. Suppose that at the second stage of the PPS design the b elements sampled from each selected PSU are drawn by SRS. Then, for the matching ultimate cluster design, within each of the population's PSUs the UCs can be formed as before, taking b elements by SRS to comprise the first UC, taking b elements by SRS from the remainder to comprise the second UC, and so on until all the elements are accounted for. In this way PSU α with B_{α} elements is divided into B_{α}/b UCs. (We assume for simplicity that B_{α}/b is an integer.) An SRS of a UCs is then equivalent to the PPS without replacement design, except that the ultimate cluster design may select more than one UC from a PSU, whereas this is not possible with the PPS design. Providing the chance of choosing two UCs from the same PSU is small with the ultimate cluster sample, the difference between the

two designs is negligible. The similarity between the two designs can be seen by noting that with the ultimate cluster design the probability of sampling a UC from PSU is proportional to the PSU's number of UCs, B_{α}/b (i.e., is proportional to the PSU's size).

Since the PPS design is epsem and has a fixed sample size, the simple sample mean

$$\bar{y}_p = \frac{\sum_{\alpha} \sum_{\beta} y_{\alpha\beta}}{n} = \frac{\sum_{\alpha} \bar{y}_{\alpha}}{a}$$

is an unbiased estimator of the population mean. The ultimate cluster sample approximation gives a variance estimator for $V(\bar{y}_p)$ from formula 16, ignoring the fpc term, as

$$v(\bar{y}_p) \doteq s_a^2/a \quad [21]$$

Also, the approximate design effect for the sample mean based on a PPS design at the first stage and SRS at the second stage is given by formula 17, namely $[1 + (b - 1)]$.

In practice, PPS sampling as described is seldom possible, because the true sizes of the sampling units are usually unknown. Often, however, good estimates are available from a recent census or some other source, and on other occasions reasonable estimates may be made by other means. Providing the estimated sizes, or measures of size, are reasonably good, their use in place of true sizes in a PPS selection procedure can serve well. It is, however, important to distinguish between the use of true sizes and estimated sizes; we will therefore reserve the term "probability proportional to size" (PPS) sampling for cases in which true sizes are employed, and will use the term "probability proportional to estimated size" (PPES) sampling for other cases. Estimated sizes, or measures of size, will be denoted by M_{α} .

Corresponding to formula 20, the selection equation for a two-stage PPES design with a PSUs selected at the first stage is

$$P(\alpha\beta) = f = \left(\frac{aM_{\alpha}}{\sum M_{\alpha}} \right) \left(\frac{b}{M_{\alpha}} \right) \quad [22]$$

An important implication of this equation is that, in order for the sample to be an epsem one, the sampling rate at the second stage is (b/M_{α}) . Applying this rate to the B_{α} elements in selected PSU gives an expected sample size from that PSU of $b(B_{\alpha}/M_{\alpha})$. This expected sample size will vary from PSU to PSU depending on the ratio (B_{α}/M_{α}) , and will equal the desired sample size b only when $M_{\alpha} = B_{\alpha}$, i.e., when the PSU's estimated size equals its true size. This variability in the sample taken from different PSUs has to be accepted to retain the epsem property; the variability will be tolerable, providing the estimated sizes are reasonably accurate.

As an illustration, suppose that the true sizes (B_{α}) of the nine blocks in the earlier example were not known. A rapid tour of the area provides some rough estimates M_{α} that are used in a PPES selection. The values of M_{α} are given below, with the expected sample sizes from each PSU, assuming that the PSU was selected. These expected sample sizes are calculated by applying the subsampling rate b/M_{α} to the B_{α} elements in the cluster taken from the earlier figures. Once selected, the true size of a PSU would be determined.

Block:	1	2	3	4	5	6	7	8	9
M_{α} :	30	110	50	20	20	50	10	50	20
B_{α} :	20	100	50	15	18	43	20	36	13
Expected sample size:	4.7	6.4	7.0	5.3	6.3	6.0	14.0	5.0	4.6

The expected sample sizes vary somewhat because of the inaccuracies in the M_{α} , but mostly the variability is reasonable. Notice, however, the large expected sample size from PSU 7 if selected, which occurs because that PSU's true size (20) has been substantially underestimated. In assigning measures of size, care needs to be taken to avoid gross underestimation, because of the difficulties created. Consider, for instance, the situation of a block with an estimated 10 dwellings based on the last census, where a recent high-rise building has just been completed with 800 new dwellings. Another notable feature of these expected sample sizes is that most of them are below the b value of 7. The explanation for

this lies in the fact that the M_{α} 's tend to overestimate the B_{α} 's: $\sum M_{\alpha} = 360$ compared with $\sum B_{\alpha} = 315$. With an intended total sample of $n = 21$, the overall sampling fraction was set at $21/360$; the expected total sample size is thus only $(21/360) 315 = 18.4$. This discrepancy draws attention to the need for a good estimate of the total population size.

As the preceding discussion shows, a consequence of using PPES sampling is that the total sample size is not fixed, but is a random variable that depends on which PSUs are selected. To emphasize this fact, the total sample size will be represented by x rather than n , and the sample mean by $r = y/x$, where y is the sample total for the y variable. The sample mean is termed a *ratio mean* or *ratio estimator* because it is a ratio of random variables. The ratio mean is not an unbiased estimator of the population mean, but the bias is negligible when the variability in x is sufficiently small. The bias can be safely ignored when the coefficient of variation of x is less than 0.1, where the coefficient of variation is defined as the standard error of x divided by its expected value, the expected sample size.

The variance of the ratio mean is complicated by the fact that its denominator is a random variable. As a result, only a large sample approximation, based on what is sometimes called the Taylor expansion or delta method, is available. The appropriate use of this approximation requires the coefficient of variation of x to be small, less than 0.2 and preferably less than 0.1. The general form of the approximate variance estimator for the ratio mean $r = y/x$ is

$$v(r) \doteq [v(y) + r^2 v(x) - 2rc(x, y)] / x^2 \quad [23]$$

where $c(x, y)$ is the sample covariance of x and y . In applying this formula, appropriate formulae need to be substituted for $v(y)$, $v(x)$, and $c(x, y)$. As an illustration, consider an epsem stratified multistage design. Let $y_{h\alpha}$ be the sample total of the y variable for PSU α in stratum h , let $x_{h\alpha}$ be the sample size in that PSU, let y_h be the total of the y variable for the a_h sampled PSUs in stratum h , and let x_h be the sample size in that stratum. Then, using the with-replacement approximation,

$$v(y) = \sum_h a_h s_{y_h}^2$$

$$v(x) = \sum_h a_h s_{xh}^2$$

$$c(x, y) = \sum_h a_h s_{xyh}$$

where

$$s_{yh}^2 = \sum_{\alpha} [y_{h\alpha} - (y_h/a_h)]^2 / (a_h - 1)$$

$$s_{xh}^2 = \sum_{\alpha} [x_{h\alpha} - (x_h/a_h)]^2 / (a_h - 1)$$

$$s_{xyh} = \sum_{\alpha} [x_{h\alpha} - (x_h/a_h)] [y_{h\alpha} - (y_h/a_h)] / (a_h - 1)$$

The generality of $v(r)$ in formula 23 with the above substitutions deserves note. The formula applies to *any* epsem stratified multistage design. It applies no matter what probabilities are used for selecting the PSUs and no matter what form of subsampling is employed within selected PSUs. The formula applies to samples in which a nonstratified selection of PSUs is taken (as the special case with only one stratum) and to PPS samples where the sample size is fixed—when $v(x) = 0$ and $c(x, y) = 0$. It can be applied to ratio means and percentages based on the total sample and on subclasses (e.g., wage earners or married persons only). The only restrictions are the need for the coefficient of variation to be less than 0.2, and the need for the with-replacement approximation to be suitable. By modifying the definitions of y_{α} and x_{α} , formula 23 can be readily extended to apply to nonepsen designs. See Kish (1965: chap. 6) for further discussion.

Before leaving the subject of PPS and PPES sampling, one further illustration will be given to draw attention to some difficulties that are often met in practice. For this purpose, the earlier example of sampling

dwelling from three blocks is modified as follows. The three blocks are now to be selected from ten with the following estimated sizes, M_{α} , with $\sum M_{\alpha} = 315$:

Block:	1	2	3	4	5	6	7	8	9	10
M_{α} :	20	120	45	15	18	43	5	0	36	13

As before, the desired sample size is 21, implying a desired subsample size of $b = 7$. There are two problems with applying the previous selection procedures to this population.

In the first place, the use of systematic sampling with an interval of 105 could lead to block 2 being selected twice. Since its size is greater than the interval, it is bound to appear once in the sample, and it has a probability of $15/105 = 1/7$ of being chosen twice: if the random start lies between 021 and 035, PSU 2 will be selected as the first selection and also as the second selection, since adding 105 to a number in this range gives a number no greater than 140, the upper number associated with block 2 in the cumulative totals. One solution is simply to accept the two selections if they occur, taking two different subsamples from the block. Another solution is to note that PSU 2 is bound to appear and is thus, in effect, a stratum. It is then set aside as a separate stratum and its elements are sampled at the overall sampling rate, here 1 in 15. Two selections are then made from the remaining blocks with PPES, also at the overall rate of 1/15. A reduction in the value of b is needed since now with 2 selections from a stratum with $M_{\alpha} = 195$, $2b/195 = 1/15$, i.e., $b = 6.5$. PSUs that are large enough to be certain to appear at least once in the sample occur very often in practice. They are frequently treated as separate strata and are termed, misleadingly, self-representing PSUs.

The overlarge PSUs are identified in the selection equation as those with first-stage selection probabilities greater than 1; in the case of block 2 its selection probability is $aM_{\alpha}/M_{\alpha} = (3 \times 120/315) = 360/315$. The other commonly encountered problem with PPS or PPES selections is undersized PSUs, which give rise to second-stage selection probabilities b/M_{α} in excess of 1. Block 7 falls in this category in the current example since $b = 6.5$ and its measure of size is only 5. One simple way to handle this problem is to link the block to a geographically adjacent one, and treat the two as one cluster. This can be done in any convenient way before the selections are made, or afterward if objective linking rules are used (see Kish, 1965: 244-245). If there are many undersized PSUs and linking would cause fieldwork difficulties, they may be placed in a